

## Supplementary Materials

### Transfer learning from computed stability data for nanobody melting-temperature prediction

Taihei Murakami, Kentaro Sasaki, Soichiro Oda, Kazuma Okada, and Yasuhiro Matsunaga

#### Supplementary methods

##### Details of the heterogeneous MD data set

The heterogeneous data set was assembled from a SABDab summary and the corresponding structures [?]. For each PDB identifier, we selected the first heavy-chain entry whose chain was present in the structure file. This rule yielded 1143 samples with a usable sequence, topology, trajectory, and native-contact label, representing 833 unique amino-acid sequences. Ninety-seven sequences occurred in more than one PDB entry and accounted for 407 samples. We calculated native-contact values separately for each structure and sampled them individually during training; duplicate sequences were not collapsed, and the model did not receive the PDB identifier. Eleven sequences exceeded the 158-residue model input and were truncated at the end.

Structures were prepared and protonated at pH 7.4 with PDBFixer, pdb2pqr, and PROPKA [?, ?]. Each protein was solvated in OPC water with 15 Å padding and 0.15 M NaCl and described with ff19SB [?, ?]. Equilibration at 300 K used 1 ns NVT and 1 ns NPT with a 2-fs timestep, after which two 1-ns restrained NVT stages at 400 K preceded 100 ns of unrestrained 400-K NVT production. Production used hydrogen-mass repartitioning to 4 amu, a 4-fs timestep, and output every 100 ps. OpenMM used PME electrostatics with a 1.0-nm cutoff, constraints on bonds to hydrogen, and a Langevin middle integrator with 1 ps<sup>-1</sup> friction [?].

The contact calculation used all backbone heavy-atom pairs separated by more than three residues and within 0.45 nm in the first production frame, as for the single mutation scans.  $Q$  was averaged over the final 30 ns of production, that is the last 300 frames of 100-ps output. The two MD data sets thus shared one  $Q$  definition, contact set, native reference, and averaging window, and differed only in the variants they covered.

##### FEP charge-change check

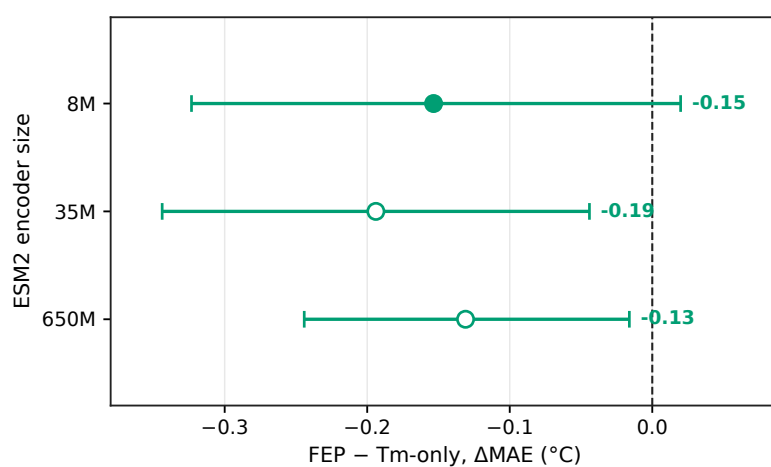
For charge-changing mutations in a cubic 70-Å box and a solvent dielectric constant of 78, we tested the leading analytical periodic net-charge (Ewald self-energy) correction [?],  $0.086(\Delta q)^2$  kcal mol<sup>-1</sup>. We applied it and repeated the sign reversal, robust scaling, Yeo–Johnson transform with standardization, and min–max scaling used for the training labels. Across the 284 charge-changing mutations the scaled labels barely moved, with a maximum shift of 0.008 on the [0, 1] scale and a correlation of 0.99989 with the uncorrected labels, so the training labels use the uncorrected  $\Delta\Delta G$ .

##### Side-chain contacts in the native-contact label

The native-contact fraction  $Q$  in the main analysis uses backbone heavy atoms only. We also evaluated the side-chain-inclusive definition, in which all protein heavy atoms are eligible for contacts. On the matched 400 K mutation scan this raised the label spread by roughly 1.4-fold, but it did not improve transfer. With hyperparameters held fixed it slightly increased the held-out  $T_m$  MAE, and re-selecting the learning rate, dropout, and encoder learning rate on the 114-example validation set did not recover the backbone result. In the fine-tuned regime the side-chain label was significantly worse than the  $T_m$ -only baseline (paired  $\Delta$ MAE +0.30 °C, 95% CI [+0.12, +0.49]), while the backbone label was neutral; in the frozen regime it erased the backbone label’s benefit ( $\Delta$ MAE from –0.20 to –0.04 °C). Because the mutant side chains were built without repacking their neighbors, side-chain contacts near each mutation carry construction noise rather than added signal, so the main analysis retains the backbone-only definition.

##### Model-size checks

We fine-tuned the 35M- and 650M-parameter ESM2 encoders without size-specific hyperparameter searches. For both sizes, the  $T_m$ -only model used the shared architecture, dropout 0.05, and an encoder learning rate of  $1 \times 10^{-4}$ . The FEP model used the shared architecture with separate 1MEL and 4IDL heads,  $n = 320$  samples per structure, dropout 0.15, and an encoder learning rate of  $3 \times 10^{-5}$ . Both conditions used a non-encoder learning rate of  $3 \times 10^{-4}$ , weight decay 0.04, and ten training seeds.



**Figure S1** The FEP result held across ESM2 encoder sizes. FEP-minus-Tm-only MAE ( $\Delta\text{MAE}$ ) with a fine-tuned encoder; negative values mean lower Tm error with FEP. The 8M point is the five-seed model selected by the two-stage search, and the 35M and 650M points are ten-seed checks without size-specific tuning. Whiskers are 95% paired bootstrap intervals over the 396 Tm test sequences.